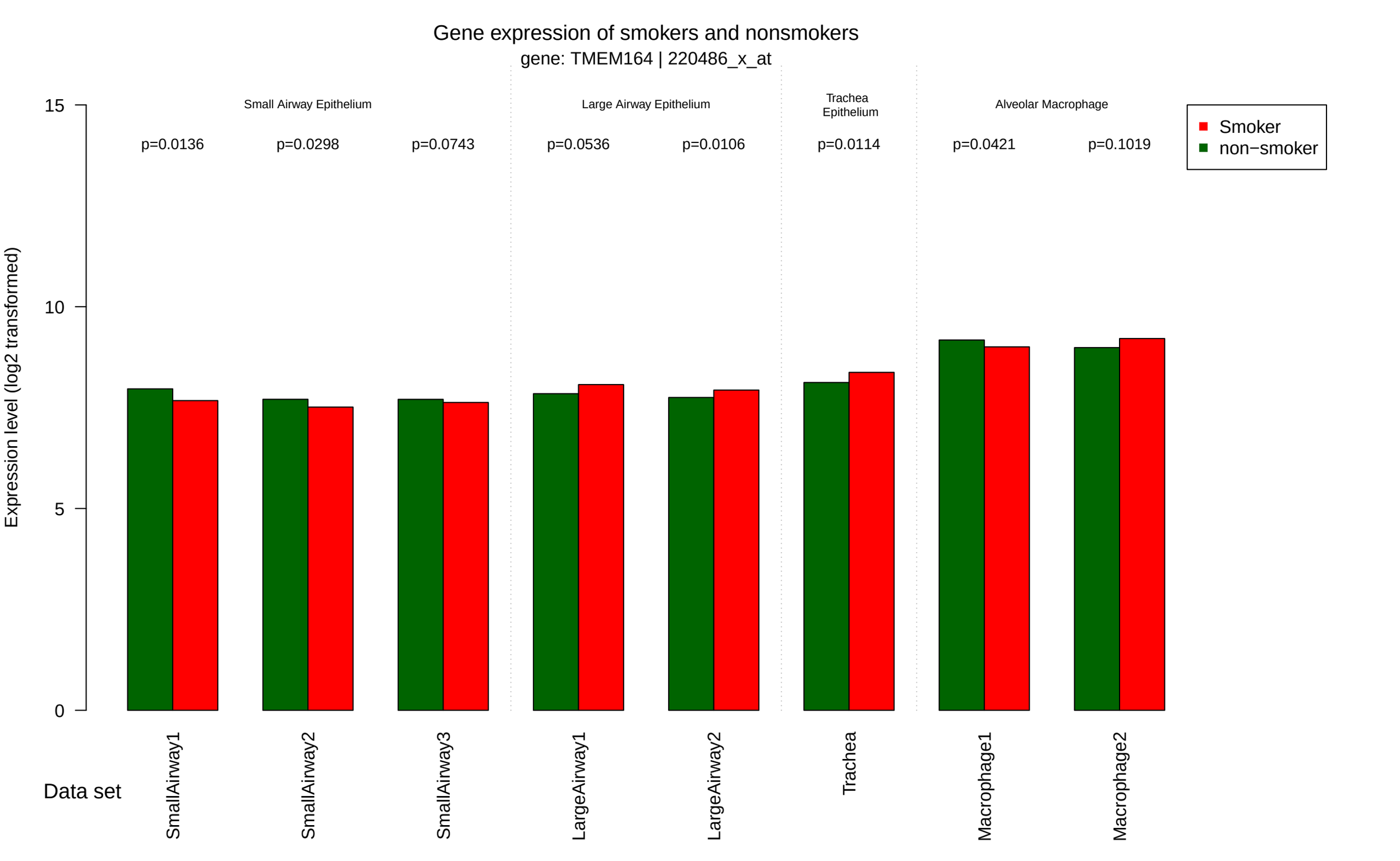
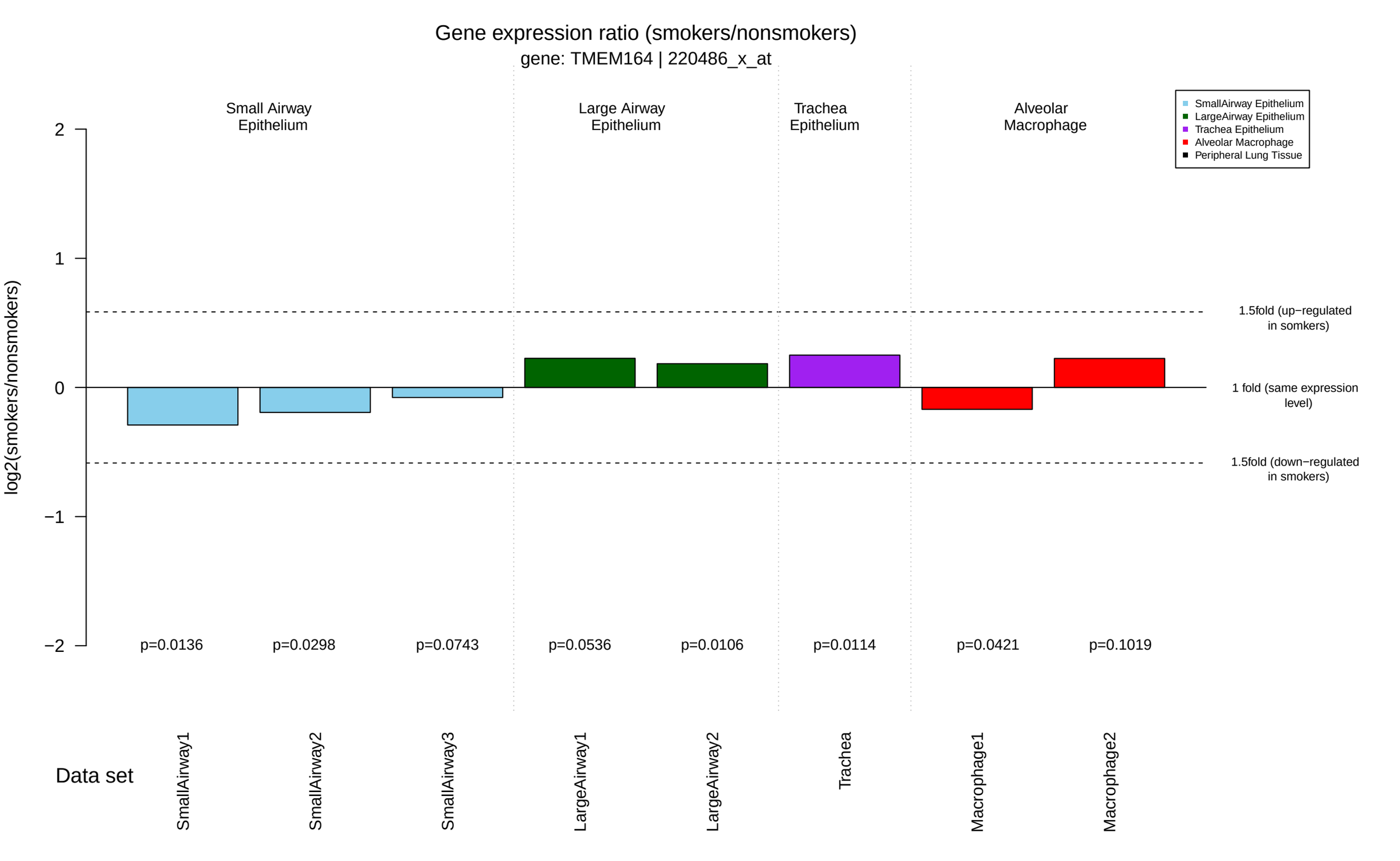
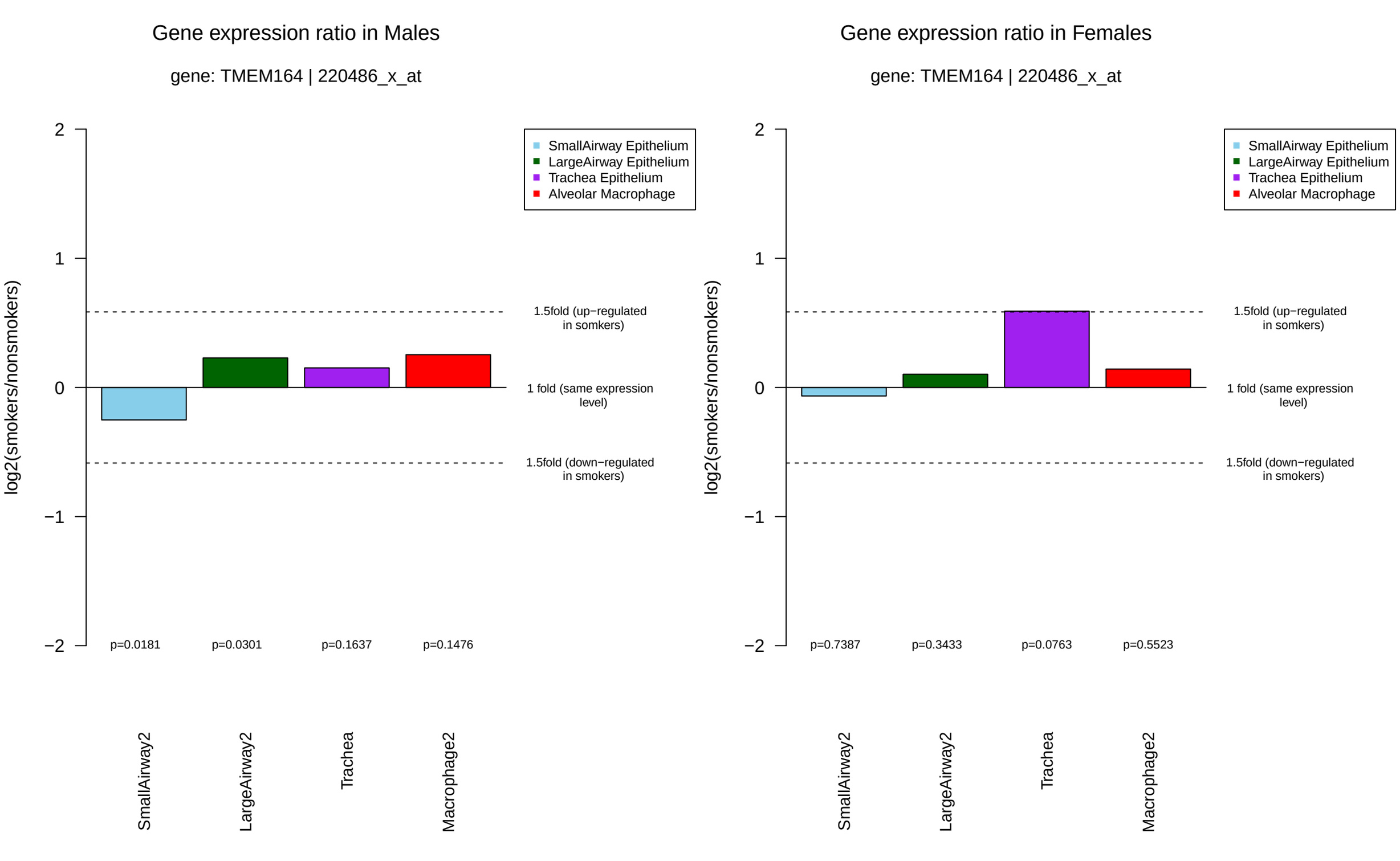
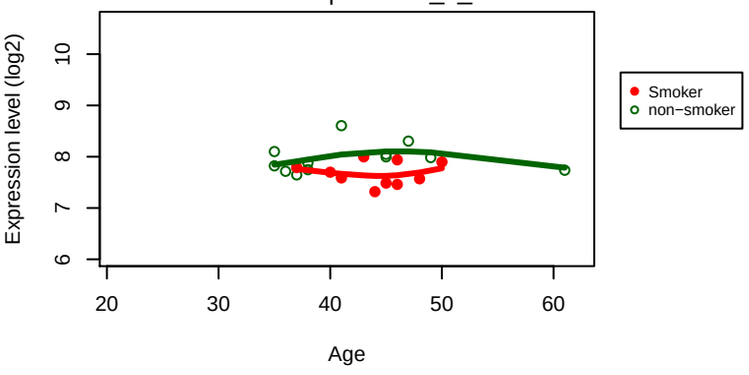
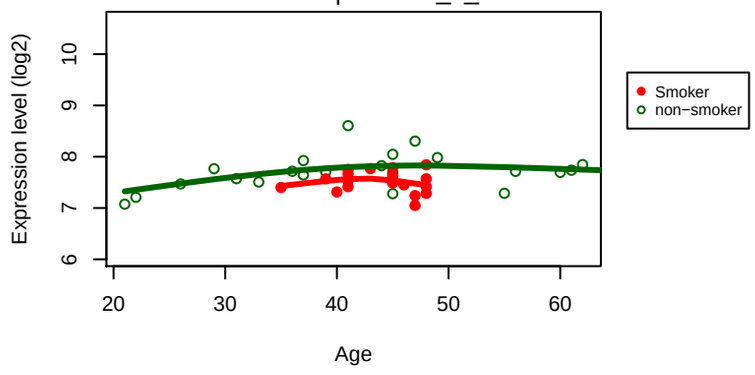
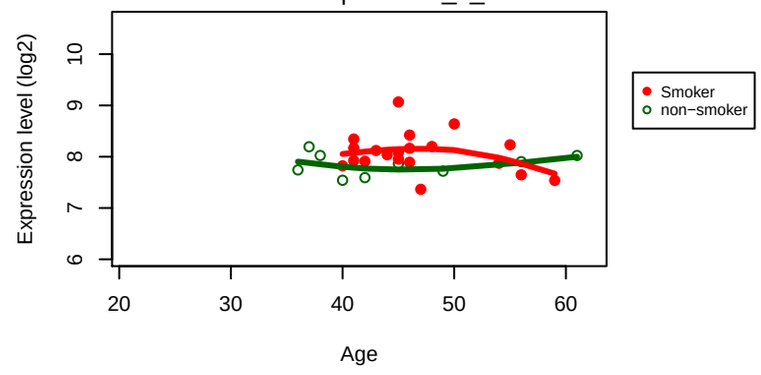

Gene expression of smokers and nonsmokers
gene: TMEM164 | 220486_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0136
p=0.0298
p=0.0743
p=0.0536
p=0.0106
p=0.0114
p=0.0421
p=0.1019
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

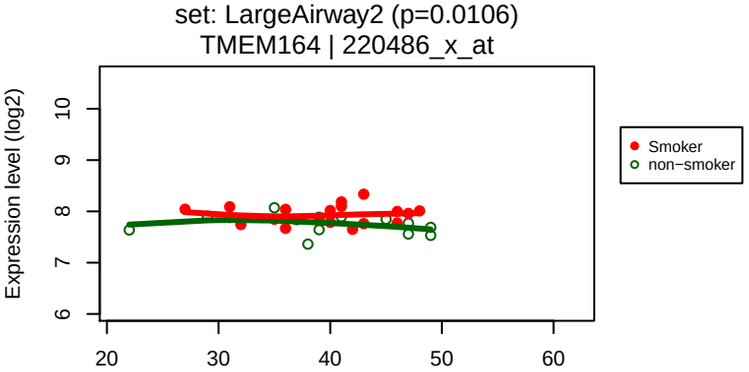

Gene expression ratio (smokers/nonsmokers)
gene: TMEM164 | 220486_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0136
p=0.0298
p=0.0743
p=0.0536
p=0.0106
p=0.0114
p=0.0421
p=0.1019
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

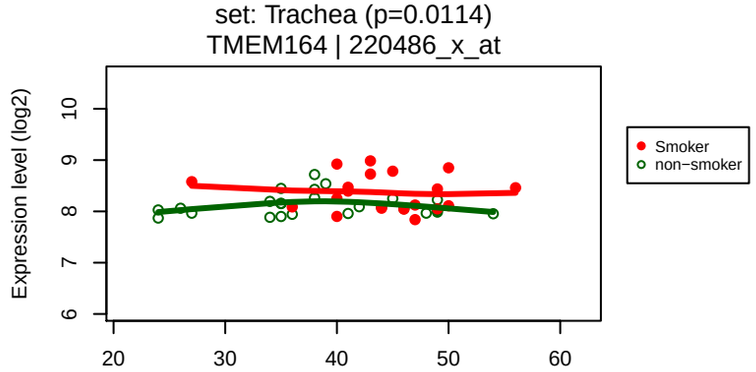

Gene expression ratio in Males
gene: TMEM164 | 220486_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0181
p=0.0301
p=0.1637
p=0.1476
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM164 | 220486_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7387
p=0.3433
p=0.0763
p=0.5523
SmallAirway2
LargeAirway2
Trachea
Macrophage2

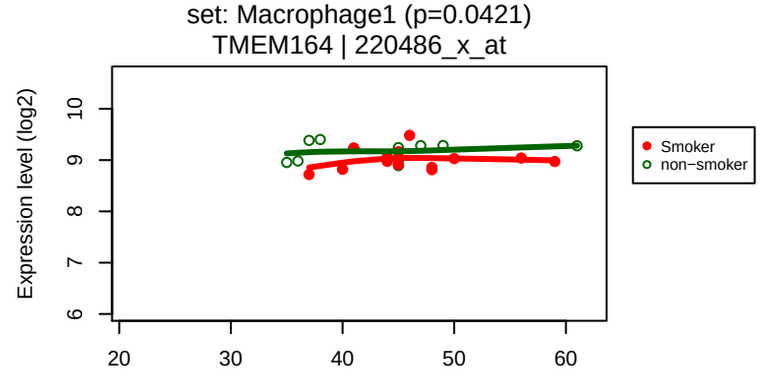

set: SmallAirway1 (p=0.0136)
TMEM164 | 220486_x_at
Expression level (log2)
Age
Smoker
non−smoker

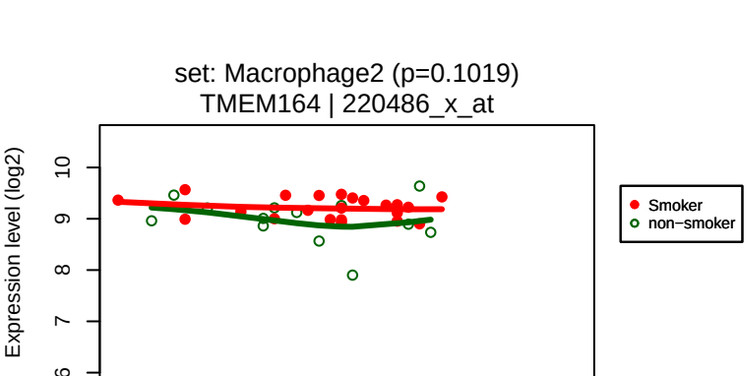

set: SmallAirway2 (p=0.0298)
TMEM164 | 220486_x_at
Expression level (log2)
Age
Smoker
non−smoker

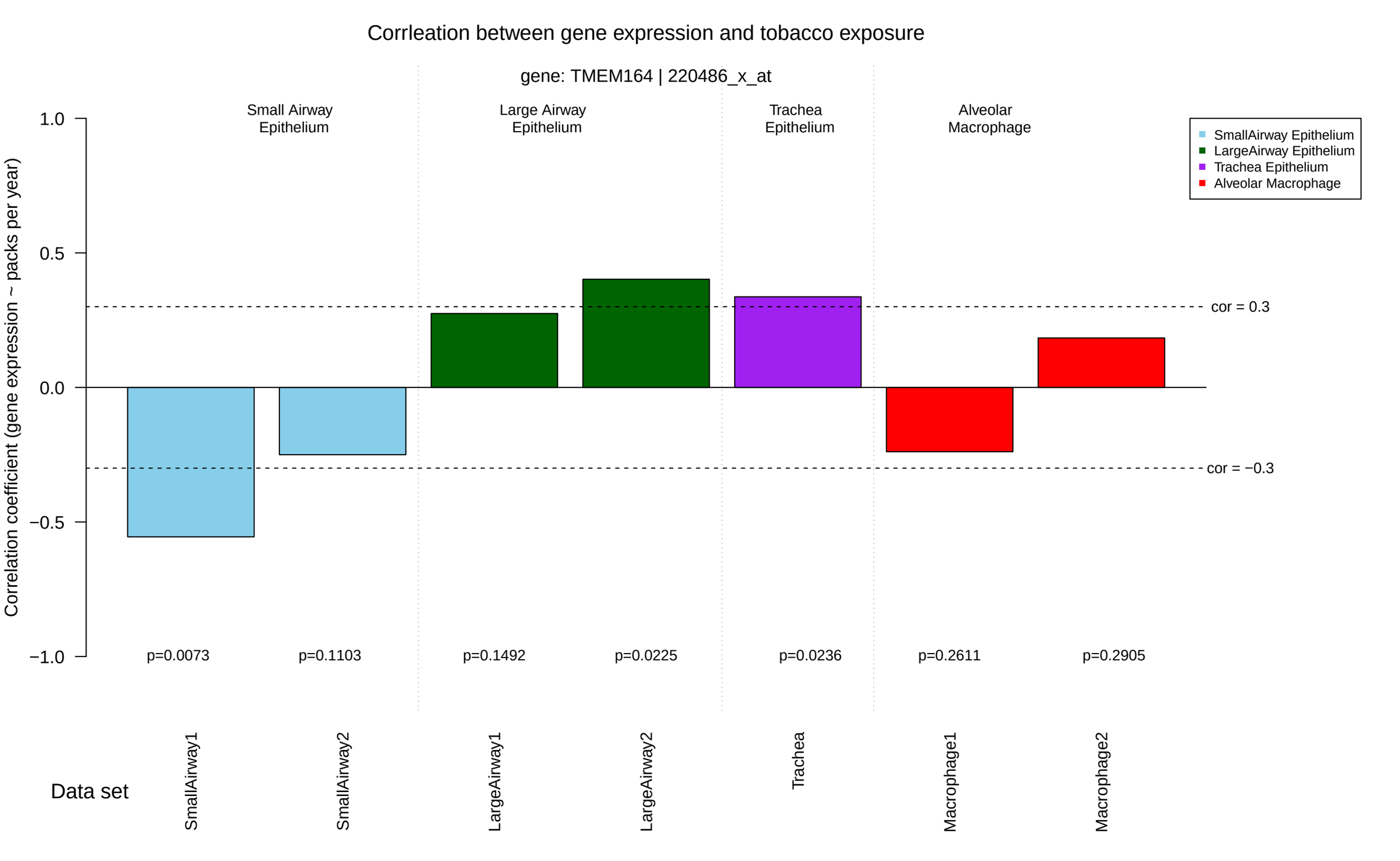

set: LargeAirway1 (p=0.0536)
TMEM164 | 220486_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0106)
TMEM164 | 220486_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0114)
TMEM164 | 220486_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0421)
TMEM164 | 220486_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1019)
TMEM164 | 220486_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM164 | 220486_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0073
p=0.1103
p=0.1492
p=0.0225
p=0.0236
p=0.2611
p=0.2905
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2